

The Staircase Heartbeat Discrimination Task (S-HDT)

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Abstract

Cardiac interoceptive accuracy has received significant attention in the growing field of interoception. Despite this, there are clear, and openly discussed limitations to the tasks commonly used to assess this dimension, including questionable construct validity, unjustified assumptions, and that, depending on the task, few participants exceed chance performance. To address this, we propose a novel version of the heartbeat discrimination task. The Staircase-Heartbeat Discrimination Task (S-HDT) addresses the aforementioned limitations and facilitates a more nuanced understanding of individual differences in cardiac interoceptive accuracy. Herein we (i) detail the experimental null, (ii) demonstrate the recoverability of known point of subjective simultaneity and accuracy values by passing simulated data through our analysis pipeline and (iii) highlight the feasibility and tolerability of this task in a young control sample ($n = 42$; mean age = 22.57 ± 4.55 ; 64.28% women). In our sample, we show the S-HDT has greater sensitivity and specificity than the original HDT, and enables the prediction of accuracy as well as individuals point of subjective simultaneity. With a running time of ~30 minutes, fully automated, open source, and freely available code (via GitHub), this task is accessible to further our understanding of cardiac interoception across multiple interoceptive domains.

Keywords:

Cardiac, interoception, interoceptive accuracy, point of subjective simultaneity

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Introduction

Interoception is broadly defined as the sensing, interpretation and integration of signals originating from within the body across both conscious and unconscious levels of processing¹, and recent models propose it can be separated into multiple different dimensions^{1,2}. One of these dimensions, interoceptive accuracy, defined as the correct and precise monitoring of afferent signals from the periphery via the central nervous system, has been a major focus of recent literature and the subject of a number of reviews^{3,4}, critiques⁵⁻⁹, and empirical studies¹⁰⁻¹⁵.

Historically, cardiac interoception has dominated the field of interoceptive accuracy, likely because heartbeats are discrete, easily measurable events. The heartbeat discrimination task (HDT) has been one of the more widely used experimental tasks to test cardiac interoceptive accuracy^{16,17}. This task requires participants to report whether an external stimulus, often a tone, is in-sync or out-of-sync with their heartbeat. In a common variant, participants are presented with tones 250ms and 550ms after the heart's R-wave, which are generally regarded to be perceived as synchronous and asynchronous, respectively¹⁸. Accuracy on this task is then typically determined by comparing the number of trials correctly identified as synchronous or asynchronous¹⁶.

While this task avoids criticism afforded to the heartbeat tracking (counting) task⁷, in that it cannot be guided by higher-order knowledge of heart rate, it is not without limitations. Firstly, more often than not, participants find this task (or its variants) particularly challenging to complete. This is reflected in performance rates above chance in only ~25% of the general population^{19,20}, which results in bimodal group distributions; those who are able to perform the task (accuracy > 75%), and those who are not (accuracy < 80%). A second limitation relates to the pre-defined intervals between the R-wave and their use in accuracy judgements²¹. Individual differences in the spatial and temporal^{20,22} perception of heartbeats may confound task performance, with prior research showing external tones can be judged as simultaneous with heartbeat sensations (i.e. an individual's point of subjective simultaneity - PSS) if presented anywhere from 0-400ms after the R-wave^{16,23-25}; the difference likely driven by the dominant mechanism by which the subjective perception of heartbeats is made (i.e. temporal perception). Betka, et al.²¹ examined this directly using a multi-interval heartbeat discrimination task (MI-HDT¹⁹), wherein a sequence of 5 tones were presented either 0, 100, 200, 300, 400 or 500ms after the R-wave and participants rated whether they perceived the tones as synchronous or asynchronous. Betka, et al.²¹ reported that while on average participants felt their heartbeat 257.40ms after the echocardiogram (ECG) R-peak, (i) the median reported synchronous delay was 300ms, (ii) single participants reported feeling their heartbeats between 100 and 400ms after the R-wave, and (iii) individuals' modal preferred interval ranged all the way from 0ms to 500ms. Furthermore, the lack of group level differences in the probability of answering "Yes" for a delay of 100ms compared to 200ms or 300ms, nor for a delay of 200ms compared to 300ms or 400ms supports the notion of considerable heterogeneity within their sample of 52 control participants. When taken together with prior work, the assumption that everyone would endorse synchrony for a tone presented at 250ms seems unsubstantiated¹⁵. While the MI-HDT has clear advantages compared to the standard version in that it presents a range of delays, the granularity is low (100ms increments) and the task takes >45 minutes to run due to the number of trials required, making it challenging for implementation in time-constrained studies.

In this manuscript, we introduce a novel version of the heartbeat discrimination task, the Staircase-Heartbeat Discrimination Task (S-HDT), that addresses the aforementioned limitations and facilitates a more nuanced understanding of individual differences in cardiac interoceptive accuracy. The S-HDT task has greater sensitivity and specificity than the original HDT, and enables the prediction of both accuracy as well as PSS, in addition, and in comparison, to the historically assumed value of 250ms.

Methods:

Staircase-Heartbeat Discrimination Task (S-HDT):

Design and Procedure:

The S-HDT has 2 phases: a training phase and a staircase phase, which take in total ~30 minutes to run.

Training Phase:

In the training phase participants practiced between 2 and 20 trials in order to ensure they understood the trial structure and how to respond appropriately. In each trial they were played 10 tones. The tones were linked to the R-wave peak of every second heartbeat and on any trial were played after a randomly selected delay between 0-600ms in 50ms increments (Figure 1). As the delay duration could reach 600ms during the staircase phase, and 750ms is the inter-R-interval of someone with 80 bpm, we postulated if the tones were played on each beat, individuals may misattribute a tone to the forthcoming heartbeat if the delay was reaching this maximal value. By playing tones on every second beat, we aimed to minimize the risk of this happening. Timing of stimulus presentation was controlled by a real-time script running on the CED-power1401 unit, identifying the QRS complex with millisecond temporal accuracy.

At the end of the tones, participants responded to “Are the beeps ‘before’, [at the] ‘same time’, or ‘after’ your heartbeat?” and provided a rating of confidence in their decision (“How confident are you?”) from 0 – Guess, to 100 – Certain. Responses were recorded by the experimenter, minimizing exteroceptive demands for the participant, and enabling them to remain with their eyes closed for the duration of the task if desired.

Staircase Phase:

The simple staircase was implemented in PsychoPy (PsychoPy 2023.2.3). This is an adaptive method used to estimate a threshold by adjusting the delay duration based on the participant's performance. The delay increases if participants submit two consecutive ‘before’ responses and decreases if participants submit two consecutive ‘after’ responses, allowing for convergence on the individual's perceptual threshold (i.e. their PSS). Responding ‘same time’ results in no movement of the staircase.

The task has 4 staircases in total, 2 that began at 400ms and 2 that began at 100ms presented in the following order: 400(A), 100(A), 400(B), 100(B). Each staircase had 15 trials, resulting in a total of 60 trials in the staircase phase of the experiment. As in the training phase, on each trial participants were played 10 tones that were played at a set delay (determined by the staircase) relative to the R-wave peak of every second heartbeat as determined by ECG. They then responded whether they perceived the tones to be ‘before’, [at the] ‘same time’, or ‘after’ their heartbeat and rated their confidence in their decision. Dependent on the participant's response (see above), the staircase determined the forthcoming delay duration. Participants were offered a short break after each staircase (after 15 trials).

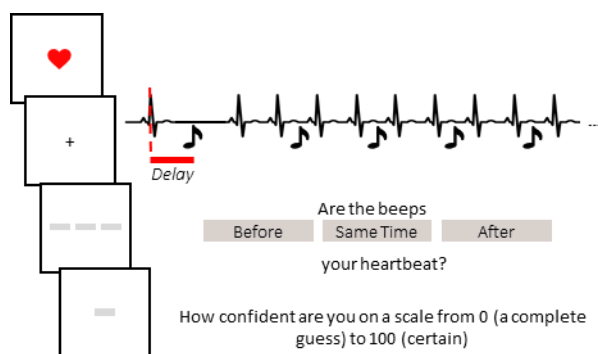


Figure 1: Trial schematic for the Staircase – Heartbeat

Discrimination Task. The start of each trial was signalled by a heart icon displayed for 2 seconds. Participants were then played 10 tones at a set delay from the R-wave peak of every second heartbeat. At the end of the tones, participants responded verbally to “Are the beeps ‘before’, [at the] ‘same time’, or ‘after’ your heartbeat?” and provided a rating of confidence in their decision (“How confident are you?”) from 0 (Guess) to 100 (Certain).

Reference Data: PSS and accuracy trajectories.

We generated 1000 simulated responses for each PSS (100-400ms in 50ms increments) and accuracy (ranging from 0-100% in 10% increments) combination for two staircases (one starting at 100ms and one starting at 400ms). As in prior relevant simulation work by others²⁶, this was achieved by randomly generating a response for each trial of the S-HDT, where the accuracy is correct relative to the PSS, proportion to the accuracy level. In other words, if the simulated accuracy level was 70%, then on any given trial there was a 70% chance of being correct (relative to the PSS) and a 30% chance of being incorrect. As Kleckner, et al.²⁶ explain, this approach captures the stochastic nature of the task and participants responses. The mean of the simulations per PSS-accuracy combination served as our reference trajectory, to which we later compare the null data, simulated, and real participants. See Figure 2A for the mean trajectories (+/- 95% confidence interval) for different reference PSS and accuracy combinations.

S-HDT analysis pipeline:

We applied a 3-step pipeline to:

- (i) identify 'random' and 'non-random' responders
- (ii) identify participants who likely applied a cognitive strategy to perform the task, and
- (iii) identify the most probable PSS and accuracy for each participant.

(i) Classify non-random and random responders:

If participants are not able to feel their heartbeat sensation, we would expect them to randomly select one of the three response options on each trial. To assess whether participants were responding randomly, we compared the frequency of each response selection to a null distribution generated from the simulated null data (see below). In other words, for each participant, we calculated the frequency of each response option (before, same time, or after) for trials from 100 (A+B) staircases, and 400 (A+B) staircases. A participant was classified as a non-random responder if the frequency of at least one of these response types for either of the staircases, fell outside the null distribution range—specifically, if it was fewer than 5 or greater than 15 times for a response option (e.g. only 3 times selecting 'before' during the 30 trials, or 17 times selecting 'after' during the 30 trials). In the simulated and real participant datasets, participants deemed to be responding randomly are excluded from further analyses.

(ii) Identify participants who used a cognitive strategy

A certain pattern of non-random, structured responses causes the trajectories move in a parallel fashion; a trajectory that rarely happens by chance, and would not happen if heartbeats are accurately detected. This pattern is indicative of participants applying a non-interoceptive, cognitive strategy when performing the task. To identify instances where this had occurred, we first averaged the data within staircase pairs (100A and B; 400A and B) to form an average 100ms staircase, and an average 400ms staircase, and then quantified the difference between the average 400ms staircase and average 100ms staircase at the final trial. If this difference had remained equal to 300ms (the difference at the start), the participant was said to have used a cognitive strategy. In the simulated and real participant datasets, these participants who used a cognitive strategy are excluded from further analyses.

(iii) Assign PSS and Accuracy level

Based on our optimization of classification accuracy for PSS values between 100-400ms, we excluded trials 1-5 from each staircase in the analysis. We concatenated the delay values from the two averaged staircases into a single vector (mean_delay_values) and calculated the root mean square error (RMSE), between the participant's vector and each PSS and accuracy combination vector from our reference data (reference_delay_values; e.g.

Figure 2B). In other words, we quantified the discrepancy between the participants data and each possible PSS and accuracy combination:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\text{mean_delay_values}_i - \text{reference_delay_value}_i)^2}$$

where n is the total number of values in the *mean_delay_value* and *reference_delay_value* arrays, and i is an index that runs from 1 to n , representing each element in the arrays.

We then used the lowest RMSE as our criterion to identify the most probable PSS and accuracy for that participant (Figure 2C – darkest blue). To determine the likelihood of obtaining these results by chance, we compared the simulated and real participants RMSE value for each the predicted PSS and accuracy to the null distribution for that PSS and accuracy. If the null distribution did not have sufficient data points to determine a distribution (i.e. less than 30) instances, the p-value was set at a conservative value of 0.001.

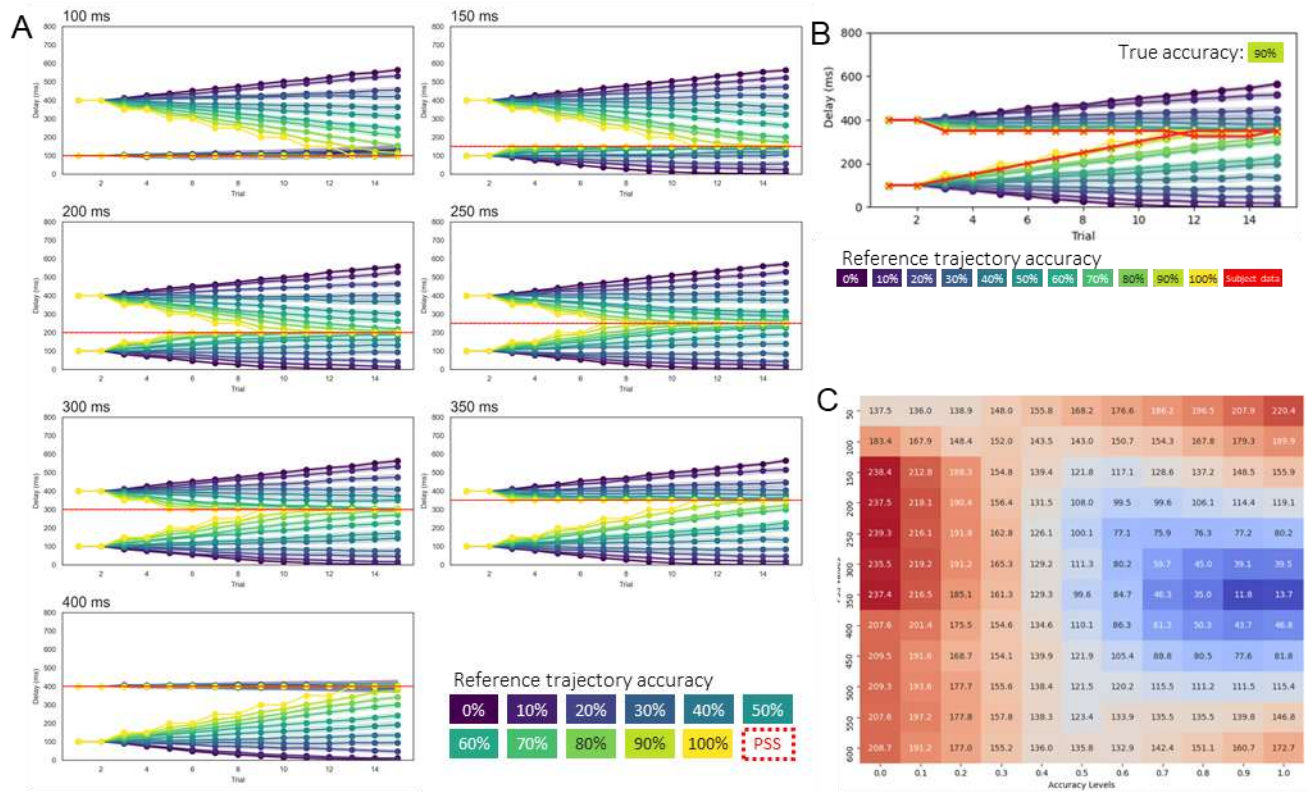


Figure 2: Staircase – Heartbeat Discrimination Task analysis pipeline to assign PSS and accuracy level. A) Reference space shows the mean trajectories (+/- 95% confidence interval) for each reference PSS and accuracy combination. B) Example of one simulated participant (true PSS = 350ms; true accuracy = 90%), overlaid on the reference trajectories for 350ms PSS. C) RMSE heatmap for one simulated participant (true PSS = 350ms; true accuracy = 90%), illustrating the minimum at PSS = 350ms and accuracy = 90%.

Application to datasets:

Simulated Null:

To create a null dataset, 10,000 participants (1000 permutations, 1000 participants) were simulated whose responses were randomized: for each trial, one of 'before', 'same time', or 'after' was chosen at random, emulating a scenario where participants' choices were not informed by cardiac interoception, but rather were selected by chance.

Null participants were parsed through the above pipeline. After completing step iii, we took the mean of the RMSE value per PSS and accuracy combination for 1000 participant fold, which generated a distribution of the mean RMSE value per PSS and accuracy combination under random conditions (1000 values; 1 mean per PSS and accuracy combination, per fold). This was then used to compare to the observed RMSE values in our simulated and real participant datasets to assess likelihood of obtaining these results, as compared to chance (the simulated null).

Simulated Participants:

To test the recoverability of known PSS and accuracy from our analysis pipeline, we simulated task data for 10,000 participants, whose true PSS and accuracy were evenly distributed across PSS values 100-400ms (in 50ms increments) and 0-100% accuracy (in 10% increments); there were 142 or 143 participants per PSS and accuracy combination. We note that it would be possible to run simulations for PSS values 50-600ms and 11 accuracy levels.

Simulated participants' data was passed through the analysis pipeline as detailed above. Participants' assigned PSS and accuracy was compared to the known (true) PSS and accuracy, to determine the proportion and characteristics of simulated participants whose true values had been successfully or unsuccessfully recovered.

Real Participants:

Sample

42 participants (mean age = 22.57 +/- 4.55; 27 (64.29%) women; 13 (30.95%) men; 2 (4.76%) non-binary individuals) were recruited from the general population via the UCL SONA recruitment platform as well as via posters, online notice boards, and word of mouth. Participants were included if they were aged between 18-65 years old and did not have/were not: a current diagnosis of epilepsy, substance use disorder, pain disorder, stroke, or diabetes; a previous or current diagnosis of schizophrenia or related conditions, or bipolar disorder; previous cardiovascular surgery; a cardiac pacemaker or implantable cardioverter defibrillator; bradycardia; a previous or current diagnosis of a significant cardiovascular or respiratory condition; currently taking antipsychotics, corticosteroids, opiates, opioids, or antidepressants – specifically NASSA, TCA, SARI or MAOI. Note the use of SSRI or SNRIs was not an exclusion criterion if the dose had not changed in the last month. Participants self-screened via an online screening form and, if eligible, were invited to attend a single 2.5-hour session at the Institute of Cognitive Neuroscience, UCL, London. All participants provided written informed consent, following a complete description of the study protocol, which was approved by the UCL Human Research Ethics Committee (ICN-VW-01-08-2024). Within this testing session, participants completed a number of interoception tasks, including a non-invasive gastrointestinal interoception task²⁷ (not referenced in the current manuscript), prior to performing, in the following order, the heartbeat tracking task, the S-HDT and the original HDT. Between these tasks they completed a number of questionnaires about emotions and mental health symptoms, which are not referenced in the current manuscript.

Physiological recording:

Physiological waveforms were recorded on PC (Power1401, Spike2 v7 software; CED). Pre-gelled adhesive Ag/AgCl electrocardiographic electrodes were used for ECG (Ambu, Denmark). ECG electrodes were placed in a modified lead III configuration, with two electrodes placed under the left and right infraclavicular fossa (just below the clavicle), and the ground electrode placed on the subject's lower left back. For the heartbeat discrimination and heartbeat tracking task, participants' heartbeats were monitored via a pulse oximeter with the sensor mounting attached to the index finger of their non-dominant hand.

Analysis

Participants' data was passed through the analysis pipeline as detailed above. Confidence was averaged across all trials. Metacognitive awareness of cardiac signals, was operationalized as the extent to which confidence predicts accuracy (per trial). To achieve this, we used the PSS derived for each participant, and re-coded each response as correct or incorrect. We then quantified insight as the area under the ROC curve of the relationship

between self-reported confidence ratings and actual performance (correct/incorrect relative to the PSS), with higher values indicating a stronger alignment between perceived and actual accuracy.

Descriptive statistics were performed to summarise subjective reports and other task parameters. Correlations were run between the S-HDT, the original HDT and the HTT.

Original Heartbeat Discrimination Task (HDT):

Participants completed 40 trials of the original heartbeat discrimination task (described in detail elsewhere¹⁸). Briefly, they were instructed “You will hear ten tones. Please can you tell me if the tones are in or out of sync with your heartbeat”. On each trial, 10 tones (100ms; 440Hz) were presented either 250ms or 550ms after the R-wave. At the end of the tones, participants responded by stating whether the tones were “in-sync” or “out-of-sync” with their heartbeats, and provided a rating of confidence in their decision (“How confident are you?”) from 0 – Guess, to 100 – Certain. Responses were recorded by the experimenter, minimizing exteroceptive demands for the participant, and enabling them to remain with their eyes closed for the duration of the task if desired.

Accuracy was quantified as the percentage of correct decisions made. Performance sensitivity was quantified using signal detection methods as d' , and bias as c' . Metacognitive awareness of cardiac signals, was operationalized as the extent to which confidence predicts accuracy, and quantified as the area under the ROC curve of the relationship between self-reported confidence ratings and actual performance.

Heartbeat Tracking Task (HTT):

Participants completed 6 trials of the heartbeat tracking task. They were instructed “Without manually checking, can you silently count each heartbeat you feel in your body from the time you hear “start” to when you hear “stop””. The time-windows over which counting occurred were: 25, 30, 35, 40, 45, and 50 seconds, presented in a randomized order. We computed the accuracy score per trial, defined as

$$\text{Trial accuracy score} = 1 - \frac{|\text{Number of heartbeats}_{\text{real}} - \text{Number of heartbeats}_{\text{reported}}|}{\left(\frac{\text{Number of heartbeats}_{\text{real}} + \text{Number of heartbeats}_{\text{reported}}}{2}\right)}$$

and averaged over trials to get an average performance score for each participant²⁸.

Metacognitive awareness of cardiac signals, was operationalized as the extent to which confidence predicts accuracy, and quantified using a correlation.

Results

Staircase – Heartbeat Discrimination Task:

Task Experience ratings:

Overall, participants found the task somewhat challenging (mean rating: 70.39 +/- 23.96; Scale: Very easy = 0 to Very hard = 100) and generally pleasant (mean rating: 52.65 +/- 22.48; Scale: Very unpleasant = 0 to Very pleasant = 100) to complete. Very few participants experienced breathlessness or had trouble breathing (mean rating: 18.68 +/- 22.77; Scale: Not at all = 0 to Very = 100). See Figure 3A.

In the training phase, participants used an average of 2 practice trials to confirm they understood the trial structure and how to respond appropriately (mean = 2 +/- 0.31; min = 2, max = 3). The delays presented in the training phase spanned the full 0-600ms range. Participants were, on average, 55.27% confident (+/- 21.38, range = 10 - 99) on these trials.

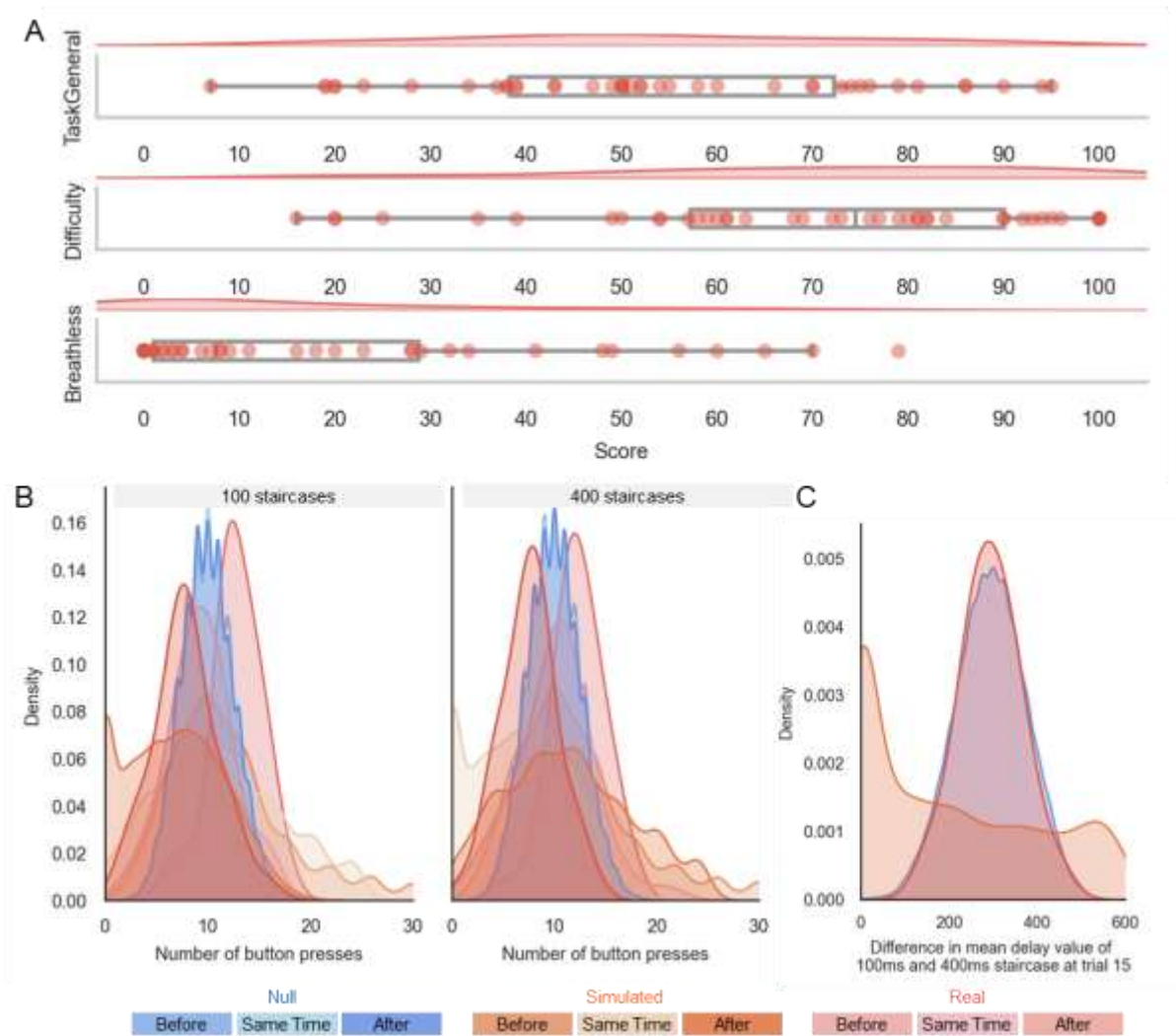


Figure 3: Identifying 'random' and 'non-random' responders and participants who likely applied a cognitive strategy to perform the task. (A) Task experience ratings. (B) Density plots show the number of button presses per response option (before, same time or after) for 10,000 participants in the simulated null sample (blues), 10,000 simulated participants (oranges), and 42 real participants (pinks). (C) The absolute difference in endpoint values (milliseconds) between the mean 100 and mean 400 staircases.

Interoceptive Accuracy:

We applied our 3-step pipeline to the simulated null dataset to establish chance performance for each step, and then to the simulated participants and our dataset of real participants.

i) Identifying 'random' and 'non-random' responders:

First, we aimed to determine how many participants from each dataset were responding randomly. In the simulated null dataset, each response option was selected an average of 10 times (± 2.58 times) per 30 trials in the 100 or 400 staircases, suggesting as would be expected, that randomly responding results in approximately one third of responses for each response option (Figure 3B). Thus, the threshold for non-random responding in the simulated or real participant samples is fewer than 5, or greater than 15 responses out of the 30 trials in each of the concatenated 100ms and/or 400ms staircases. No simulated null participants were excluded. This occurred in 1522 (15.22%) of simulated null participants, wherein 788 simulated null participants (7.88%) had pressed a single response button >5 or <15 for the 100ms staircase, and 792 simulated null participants (7.92%) for the 400ms staircase. Extreme frequencies co-occurring in both staircases occurred in only 58 simulated null participants (0.58%). 8173 (87.31%) simulated participants and 25 (59.52%) real participants had response frequencies outside the null distribution, or in other words,

for at least one of the staircase pairs they used one of the response buttons fewer than 5 or greater than 15 times (Figure 3B). Participants who did not meet this criteria were excluded from further analysis ($n = 1827$ (18.27%) simulated participants; 17 (40.47%) real participants).

ii) Identifying participants who likely applied a cognitive strategy to perform the task:

Second, in an effort to identify structured, non-random responses, we determined how many participants mean staircases moved in a parallel fashion. 942 (9.42%) of 10,000 simulated null participants had trajectories that ended exactly 300ms apart (mean final delay of 400ms staircase = 402.17ms; mean final delay of 100ms staircase = 102.17ms). No simulated null participants were excluded. 164 (1.64%) simulated participants had trajectories that ended exactly 300ms apart (mean final delay of 400ms staircase = 407.92ms; mean final delay of 100ms staircase = 107.92ms). 3 (7.14%) real participants had trajectories that ended exactly 300ms apart (mean final delay of 400ms staircase = 400ms; mean final delay of 100ms staircase = 100ms) suggesting they had used a cognitive strategy to perform the task. These participants were excluded from further analysis. 22 real participants (52.38%) therefore have data that can be further interpreted as non-random and unlikely to occur by chance.

iii) Identifying the most probable PSS and accuracy for each participant:

The null distribution of assigned PSS and accuracy labels was determined and is illustrated in Figure 4A. As expected, accuracy labels ranging from 20% to 40% were observed across all PSS values. For PSS values of 100ms and 400ms, accuracy labels were also frequently below 20% or within the 50–60% range. This pattern occurs because, in practice, the RMSE calculation is more strongly influenced by the moving staircase (as opposed to the staircase that remains around the start delay) when the PSS values match the start position of one of the staircases (Supplementary Figure 1).

Our method was able to identify both the true PSS and accuracy level in 2807 (28.54%) simulated participants, which was most achievable when the true accuracy level was less than 10% or greater than 60% (Figure 4B and C; Supplementary Figure 2). To account for the overlap in the distribution of PSS $\times$ Accuracy outcomes, we assessed success against two more lenient thresholds for evaluating prediction accuracy (Figure 4C): if the predicted PSS and predicted accuracy level was within 50ms and 10%, or within 100ms and 20%, respectively. The predicted PSS and accuracy level were within 50ms and 10% of the true value in 6194 (62.97%) of participants, and within 100ms and 20% in 7834 (79.65%) of participants.

Irrespective of assigned accuracy level, our method was able to identify the correct PSS value in 5256 (53.44%) of simulated participants. 6913 (70.28%) participants had their predicted PSS values within 50ms of the true PSS value, and 7938 (80.7%) of simulated participants had a predicted PSS values within 100ms of the true PSS value. The mean difference between the predicted PSS and true PSS was -0.08ms (+/- 100.99ms).

Accuracy level, irrespective of PSS, was correctly identified in 4542 (46.18%) of simulated participants. 7774 (79.04%) of simulated participants had a predicted accuracy value within 10% of the true accuracy level, and 9281 (94.36%) of simulated participants had predicted accuracy level within 20% of the true accuracy value. The mean difference between the predicted accuracy and true accuracy was -1.21% (+/- 12.07%).

The frequency of participants' PSS and accuracy is illustrated in Figure 4D, and assignments per subject can be found in Supplementary Table 1.

Interoceptive Confidence:

Participants' mean confidence in their responses was not significantly correlated with their accuracy ($r = -0.12$, $p = 0.49$), nor their PSS value ($r = -0.0044$, $p = 0.98$). Participants' mean confidence in their responses was significantly negatively correlated with perceived task difficulty ($r = -0.27$, $r^2 = 0.074$, $p = 2.54 \times 10^{-6}$), such that the people who found the task more difficult, were overall less confident throughout the task.

Interoceptive Metacognition: Participants mean AUC was 0.47 +/- 0.07 (min = 0.31, max = 0.63).

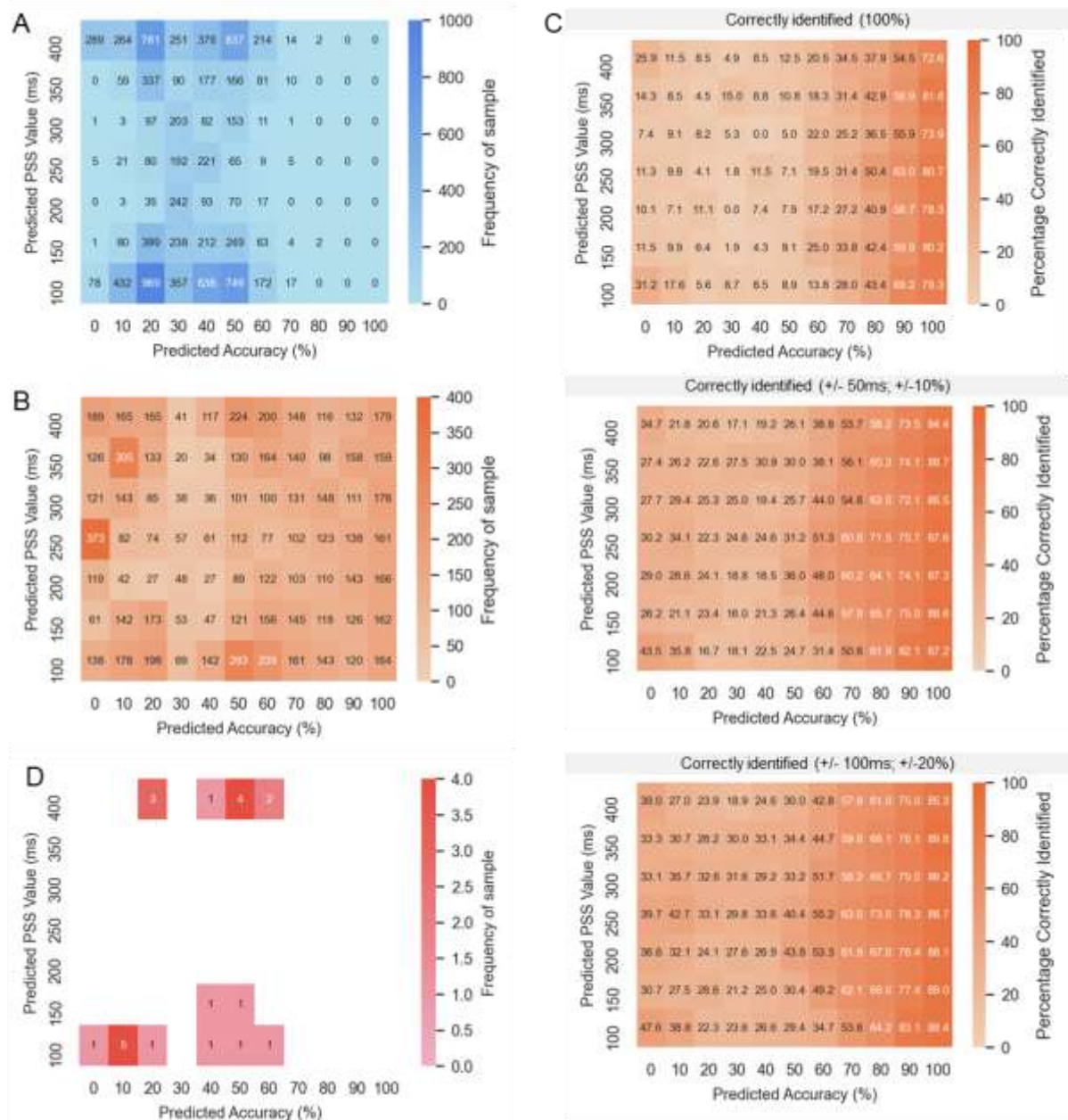


Figure 4: Identifying the most probable PSS and accuracy for each participant (A) Number of individuals for each predicted PSS and accuracy level combination, for 1000 permutations of 1000 participants. (B) Number of individuals for each predicted PSS and accuracy combination, for 10,000 simulated participants. (C) Percentage of simulated participants for whom the predicted PSS and accuracy level were correct at three thresholds – 100%; within 50ms and 10%; and within 100ms and 20%. (D) Number of real participants for each predicted PSS and accuracy level combination.

HDT:

Participants' accuracy ranged from 37% to 77% on the HDT task (mean = 56.88 +/- 10.71%). 7 of the 42 participants' (17.08%) accuracy was greater than chance (68%), but only 1 participant (2.43%) was more than 80% accurate. Mean c' was 0.58 +/- 0.16 (range = 0.19 – 0.90), and response bias varied across the sample (mean d' = 0.18 +/- 0.28, range = -0.22 – 0.89). Accuracy was not significantly correlated with confidence (r = 0.281, p = 0.07).

HTT:

Participants' scores ranged from -0.49 to 0.96 on the HTT task (mean = 0.46 +/- 0.34). 8 of the 42 participants' (19.51%) score was greater than 0.8. Accuracy was not significantly correlated with confidence (r = 0.29, p = 0.083).

Relation between S-HDT, HDT and HTT:

Accuracy:

Accuracy scores on the staircase HDT were not correlated to that on the original HDT (r = -0.22, p = 0.178), nor the HTT (r = -0.22, p = 0.064). Accuracy on the original HDT was also not related to that on the HTT (r = 0.30, p = 0.055).

Confidence:

Mean confidence ratings on the staircase HDT were strongly correlated to those on the original HDT (r = 0.88, p = 1.12×10^{-13}), as well as the HTT (r = 0.58, p = 9.59×10^{-5}). Similarly, confidence on the original HDT was related to that on the HTT (r = 0.56, p = 0.0001).

Discussion:

We introduce a novel version of the HDT – the staircase heartbeat discrimination task (S-HDT). This task uses an adaptive staircase procedure to determine an individual's most likely PSS (ms post R peak) and accuracy (%). We present data on the recoverability of a known PSS and accuracy from simulated participants, before applying the analysis pipeline to a sample of 42, young control participants. This demonstrates that the S-HDT enables identification of individuals who respond non-randomly and who likely use a cognitive strategy to perform the task, as well as estimation of participant's PSS and accuracy percentages.

Interpretation of the original HDT conflates chance responding and accuracy, whereby responses greater than 80% accurate are said to be unlikely to occur by chance; only one individual in our sample achieved this. In contrast to the original HDT which unjustifiably assumes a fixed PSS¹⁵ and provides only an accuracy level, and to the MI-HDT which assign a PSS based on the frequency of responses but only provides a binary 'detector'/'not-detector' classification, using the S-HDT we can separate these two response components to provide both a PSS and accuracy for each participant, which can be interpreted within the context of the probability of obtaining that PSS and accuracy by chance. Response frequencies significantly different from chance occurred in 48.41% of participants who performed the S-HDT. This suggests that for these individuals, we can meaningfully interpret the derived accuracy measures as reflective of an underlying strategy, most likely driven by interoceptive mechanisms. This decreases the likelihood of obtaining a bi-modal distribution, as is often seen when using the original HDT, and means that we can interpret data from individuals who may have low accuracy levels. The S-HDT thus provides a more sensitive measure of non-random, or structured responding (unrelated to a fixed accuracy level), and means the responses of a greater proportion of participants can be further interpreted as meaningful patterns of behaviour. We believe the measure of accuracy obtained by our task is a better reflection of a participant's true ability, due to the more stringent checks for chance performance.

Participants' confidence ratings on the S-HDT and the original HDT were very similar and highly positively correlated, suggesting that the experience of completing the tasks is comparable; had the S-HDT been much harder to do, it would likely have made them less confident, and this relationship would not emerge. Similarly, insight was comparably lacking across both tasks, suggesting that neither task was more accessible to higher-order awareness than the other. This may be due to the inherent challenge of sensing heartbeats at rest. Future implementations of these tasks may use interventions such as physical exercise to increase the subjective awareness of the afferent signal, with the aim of magnifying individual differences in performance. This is also relevant to probing the higher-order interpretation of cardiac interoceptive signals, which is, arguably, only consciously performed when signals reach conscious awareness. Further, evidence suggests the relation of the bodily sensations and subjective experience, including symptoms of psychopathology, is highly related to the level of insight an individual has²⁹, which is then open for (mis-)interpretation regarding the cause and meaning of these sensations.

The novel task we have designed addresses many recent recommendations to the field of interoception regarding measuring interoceptive accuracy³. Good task performance is highly unlikely to be achieved by chance, and our analysis pipeline is able to disentangle participants who likely used a cognitive strategy and those responding randomly, with increased sensitivity compared to the original HDT. Task performance is not contaminated by memory capacity - as far as participants are aware each trial is distinct from the previous one - and they are not required to compare between two intervals, as on a 2-interval forced choice design, nor to continuously update their working memory as in the heartbeat tracking task. While the task is quite challenging to perform, participants ratings of difficulty and general task pleasantness suggest the task is well tolerated. A clear and known limitation, true to the original HDT and this novel adaptation, is the requirement for multi-modal processing. In comparison to the MI-HDT which, in practice takes 45-90 minutes, the S-HDT can be completed in ~30 minutes, including a brief training and 3 short breaks. With a total of 60 trials – 15 trials per 4 staircases – this design is optimised for both sustained performance³⁰ and reliability²⁶, while ensuring feasibility in future studies.

This task could easily be adapted to present stimuli other than tones at the delay period, such as visual or tactile stimulation, which would facilitate exploration into the differences in PSS timing between these different modalities both within and between subjects. Additional staircases could be added to determine the influence of added conditions, such as a breath-hold condition³¹, and the intermittent breaks also allow for the easy addition of interventions, such as repeated bouts of physical exercise, which we hypothesise could enhance performance on the task overall and potentially facilitate heartbeat detection in individuals who otherwise do not feel these sensations at rest. The staircase method could be adapted to include a modulator of response weight to control for the underlying signal intensity, such as cardiac output. Future studies should further validate the test-retest reliability of this task, as well as the influence of homeostatic state and personality traits on performance.

In this manuscript, we introduce a novel version of the heartbeat discrimination task, the S-HDT, that addresses the limitations of the original HDT and the multi-interval HDT, and facilitates a more nuanced understanding of individual differences in cardiac interoception. We demonstrate that this task has greater sensitivity and specificity than the original HDT, as it enables the prediction of both accuracy as well as PSS. With a running time of ~30 minutes, fully automated, open source, and freely available code (via GitHub), we anticipate this task can further our understanding of cardiac interoception.

Code availability:

All code to run the task is available at: <https://github.com/Hannah-Savage/Staircase-Heartbeat-Discrimination-Task>. Here you will also find installation instructions, links to the equipment used, and comprehensive instructions to run the task. Code to perform the analysis in this manuscript will also be made available.

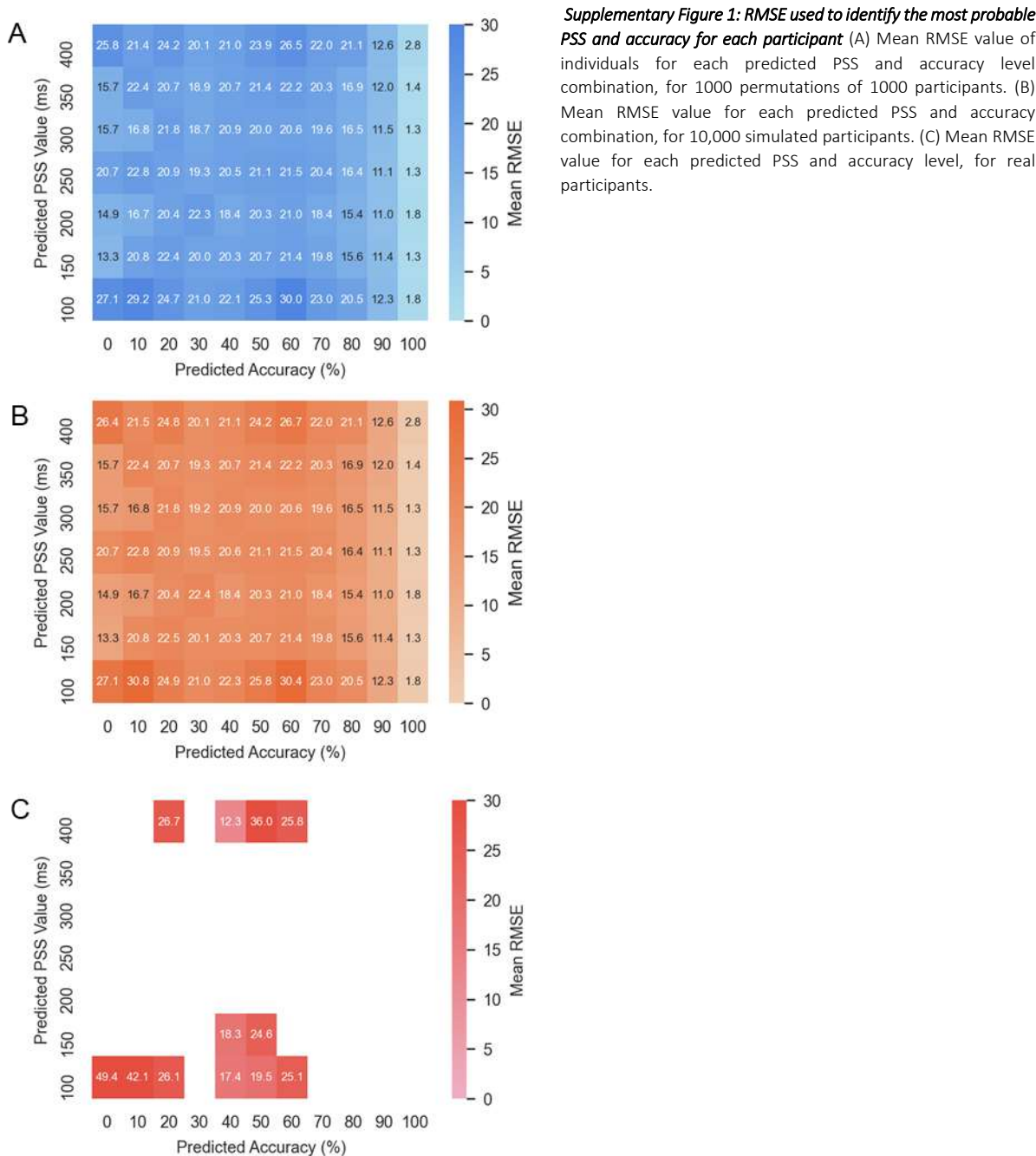
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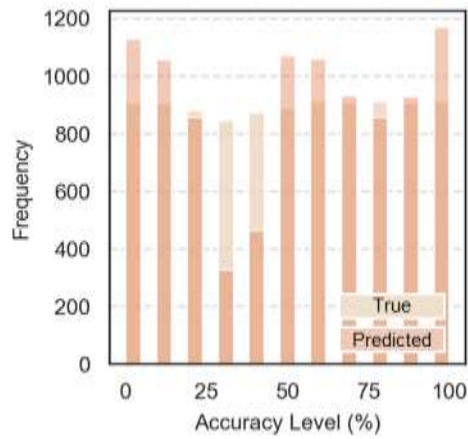
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The Staircase – Heartbeat Discrimination Task (S-HDT)

Supplementary Information





Supplementary Figure 2: Frequency of simulated participants in each Accuracy level for True Accuracy level and Predicted Accuracy Level. Accuracy levels of 30% and 40% were often under-predicted. This is in part due to exclusions of random button pressing which would disproportionately remove simulated participants from this accuracy level which is approximately equal to chance.

Supplementary Table 1: Participant's predicted point of subjective simultaneity (PSS) and accuracy. Root mean square error (RMSE) between the participant's vector and the winning PSS and accuracy combination vector from our reference data. P-value represents the likelihood of obtaining this result, as compared to the null distribution for that PSS and accuracy combination.

Participant ID	Predicted PSS Value	Predicted Accuracy	RMSE	p value
1	400	50.00%	30.907	0.994
2	100	40.00%	17.435	0.034
3	100	20.00%	26.145	0.871
4	400	50.00%	36.024	1.000
5	400	20.00%	19.416	0.041
6	100	50.00%	19.496	0.006
7	400	50.00%	44.077	1.000
8	100	10.00%	21.182	0.071
9	100	10.00%	48.925	1.000
10	100	10.00%	22.331	0.105
11	100	60.00%	25.112	0.228
12	400	50.00%	33.000	1.000
13	100	0.00%	49.393	1.000
14	400	60.00%	32.167	0.694
15	150	40.00%	18.328	0.400
16	150	50.00%	24.591	0.907
17	400	40.00%	12.257	0.001
18	100	10.00%	41.892	0.992
19	400	60.00%	19.422	0.136
20	400	20.00%	42.962	1.000
21	100	10.00%	76.092	1.000
22	400	20.00%	17.699	0.010